

Package ‘EPASnowModel’

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Type Package

Title EPA Snow Model

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Description

Process based snow models (PBSM) with two variables or seven variables (daily and subdaily)
Six snow models with daily data: Original PBSM, PBSM with S-curve, PBSM with S-curve and sublimation,

PBSM with 7 variables, PBSM with 7 variables and surface sublimation and PBSM with 7 variables and surface/blowing sublimation

A snow model with subdaily data: PBSM_EPA with 6 subdaily meteorological variables

License What license is it under?

Encoding UTF-8

LazyData true

RoxygenNote 7.3.1

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AtmosphericEmissivity *AtmosphericEmissivity*

Description

AtmosphericEmissivity

Usage

AtmosphericEmissivity(airtemp, cloudiness, vp = NULL, opt = "linear")

Arguments

airtemp	Air temperature in degree Celcius
cloudiness	Cloudiness in fraction
vp	Vapor pressure
opt	option (default=linear)

Value

Atmospheric Emissivity

declination *solar declination*

Description

solar declination

Usage

declination(Jday)

Arguments

Jday	Julian date or day of the year [day]
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Value

solar declination [rad]

EstCloudiness	<i>Cloudiness</i>
Description	
Cloudiness	
Usage	
<pre>EstCloudiness(Tx = (-999), Tn = (-999), trans = NULL, transMin = 0.15, transMax = 0.75, opt = "linear")</pre>	
Arguments	
Tx	Daily Maximum temperature in degree Celcius
Tn	Daily Minimum temperature in degree Celcius
trans	transmissivity (default=NULL-> use transmissivity function)
transMin	Minimum transmissivity (default=0.15)
transMax	Maximum transmissivity (default=0.75)
opt	Cloudiness estimation methods: Black or linear (default=linear)
Value	
Cloudiness in fraction	

Longwave	<i>Longwave</i>
Description	
Incoming longwave radiation based on the Sephan-Boltzman equation	
Usage	
Longwave(emissivity, temp, TimeStep = 24)	
Arguments	
emissivity	Terrestrial Emissivity
temp	temperature of the emitting body in degree Celsius
TimeStep	Time step in hour: Ex) 1 hour=1, daily=24 (default)
Value	
Incoming longwave radiation [kJ m-2/TimeStep]	

PotentialSolar	<i>Potential Solar</i>
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Description

Extraterrestrial (potential) solar radiation at the edge of the atmospher

Usage

PotentialSolar(lat, Jday)

Arguments

Jday

Value

Extraterrestrial (potential) solar radiation [kJ m-2 d-1]

SatVaporDensity	<i>Saturated Vapor Density</i>
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Description

Saturated Vapor Density

Usage

SatVaporDensity(T_C)

Arguments

T_C Temperature in degree Celsius

Value

Saturated Vapor Density [kg/m3]

Examples

SatVaporDensity(5)

slopefactor	<i>Slope factor</i>
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Description

Adjust solar radiation for land slope and aspect relative to the sun, 1=level ground

Usage

```
slopefactor(lat, Jday, slope, aspect)
```

Arguments

lat	Latitude [rad]
Jday	Julian date or day of the year [day]
slope	Slope of the ground [rad]
aspect	Ground aspect [rad from north]

Value

Slope factor [0,1]

SnowMelt_EPA	<i>EPA PBSM snow model</i>
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Description

Process-based snow model with seven hydroclimate variables (prcp, min/max temperature, wind speed, relative humidity and short/longwave radiations)

Usage

```
SnowMelt_EPA(  
  Date,  
  precip_mm,  
  Tmax_C,  
  Tmin_C,  
  windSp_ms,  
  Relhum_frac,  
  swrad_wm2,  
  lwrad_wm2,  
  lat_deg,  
  slope = 0,  
  aspect = 0,  
  tempHt = 1,  
  windHt = 2,  
  groundAlbedo = 0.25,  
  SurfEmissiv = 0.95,  
  forest = 0,
```

```

    startingSnowDepth_m = 0,
    startingSnowDensity_kg_m3 = 550,
    Train = 10,
    Tsnow = 0,
    Max_SnowDensity_kg_m3 = 550
)

```

Arguments

Date	Date(YYYY-MM-DD)
precip_mm	Precipitation in mm/day
Tmax_C	Maximum temperature in degree Celsius
Tmin_C	Minimum temperature in degree Celsius
windSp_ms	Wind speed in m/sec
Relhum_frac	Relative humidity in fraction
swrad_wm2	Incoming shortwave (solar) radiation in W/m2
lwrad_wm2	Incoming longwave radiation in W/m2
lat_deg	Latitude in degree
slope	Slope factor
aspect	Ground aspect [rad from north]
tempHt	Height of temperature station
windHt	Height of wind station
groundAlbedo	Ground albedo (default=0.25)
SurfEmissiv	Surface Emissivity
forest	Forest ratio
startingSnowDepth_m	Initial value of snow depth in meter
startingSnowDensity_kg_m3	Initial value of snow density in kg/m3
Train	Upper temperature of S curve: precipitation is all rain if temperature >= Train
Tsnow	Lower temperature of S curve: precipitation is all snow if temperature <= Tsnow
Max_SnowDensity_kg_m3	Maximum snow density in kg/m3, default=550 from Voordendag et al.(2021)

Details

A modified version of PBSM-S-Curve which simulates snow processes with seven hydroclimate variables. All of hydroclimate variables are from new version of the hybrid climate dataset.

Value

Time series of snow simulation

SnowMelt_EPA_Blow_Sublim

EPA PBSM snow model with blowing sublimation and Surface Sublimation

Description

Process-based snow model with blowing/surface sublimation and seven hydroclimate variables (prcp, min/max temperature, wind speed, relative humidity and short/longwave radiations)

Usage

```
SnowMelt_EPA_Blow_Sublim(
  Date,
  precip_mm,
  Tmax_C,
  Tmin_C,
  windSp_ms,
  Relhum_frac,
  swrad_wm2,
  lwrad_wm2,
  lat_deg,
  slope = 0,
  aspect = 0,
  tempHt = 1,
  windHt = 2,
  groundAlbedo = 0.25,
  SurfEmissiv = 0.95,
  forest = 0,
  startingSnowDepth_m = 0,
  startingSnowDensity_kg_m3 = 550,
  Train = 10,
  Tsnow = 0,
  Max_SnowDensity_kg_m3 = 550
)
```

Arguments

Date	Date(YYYY-MM-DD)
precip_mm	Precipitation in mm/day
Tmax_C	Maximum temperature in degree Celsius
Tmin_C	Minimum temperature in degree Celsius
windSp_ms	Wind speed in m/sec
Relhum_frac	Relative humidity in fraction
swrad_wm2	Incoming shortwave (solar) radiation in W/m2
lwrad_wm2	Incoming longwave radiation in W/m2
lat_deg	Latitude in degree
slope	Slope factor

aspect	Ground aspect [rad from north]
tempHt	Height of temperature station
windHt	Height of wind station
groundAlbedo	Ground albedo (default=0.25)
SurfEmissiv	Surface Emissivity
forest	Forest ratio
startingSnowDepth_m	Initial value of snow depth in meter
startingSnowDensity_kg_m3	Initial value of snow density in kg/m3
Train	Upper temperature of S curve: precipitation is all rain if temperature $\geq$ Train
Tsnow	Lower temperature of S curve: precipitation is all snow if temperature $\leq$ Tsnow
Max_SnowDensity_kg_m3	Maximum snow density in kg/m3, default=550 from Voordendag et al.(2021)

Details

A modified version of PBSM-S-Curve which simulates snow processes with seven hydroclimate variables. All of hydroclimate variables are from new version of the hybrid climate dataset.

Value

Time series of snow simulation

SnowMelt_EPA_Subdaily *EPA PBSM snow model with sub-daily input data*

Description

Process-based snow model with six hydroclimate variables (prcp, air temperature, wind speed, relative humidity and short/longwave radiations) at a sub-daily time step (e.g., hourly data)

Usage

```
SnowMelt_EPA_Subdaily(
  Date,
  time.step = 1,
  precip_mm,
  Tair,
  windSp_ms,
  Relhum_frac,
  swrad_wm2,
  lwrad_wm2,
  lat_deg,
  tempHt = 1,
  windHt = 2,
  groundAlbedo = 0.25,
  SurfEmissiv = 0.95,
  forest = 0,
```



```

    startingSnowDepth_m = 0,
    startingSnowDensity_kg_m3 = 550,
    Max_SnowDensity_kg_m3 = 550
)

```

Arguments

Date	Date(YYYY-MM-DD HH:MM)
time.step	Time.step in hours (default=1)
precip_mm	Precipitation in mm/day
Tair	Air temperature in degree Celsius
windSp_ms	Wind speed in m/sec
Relhum_frac	Relative humidity in fraction
swrad_wm2	Incoming shortwave (solar) radiation in W/m2
lwrad_wm2	Incoming longwave radiation in W/m2
lat_deg	Latitude in degree
tempHt	Height of temperature station
windHt	Height of wind station
groundAlbedo	Ground albedo (default=0.25)
SurfEmissiv	Surface Emissivity
forest	Forest ratio
startingSnowDepth_m	Initial value of snow depth in meter
startingSnowDensity_kg_m3	Initial value of snow density in kg/m3
Max_SnowDensity_kg_m3	Maximum snow density in kg/m3, default=550 from Voordendag et al.(2021)

Details

A modified version of PBSM which simulates snow processes with seven hydroclimate variables at sub-daily time step. All of hydroclimate variables are usually from weather forecast data (e.g., GEPS and CFS).

Value

Time series of snow simulation

SnowMelt_EPA_Sublim *EPA PBSM snow model*

Description

Process-based snow model with surface sublimation and seven hydroclimate variables (prcp, min/max temperature, wind speed, relative humidity and short/longwave radiations)

Usage

```
SnowMelt_EPA_Sublim(
  Date,
  precip_mm,
  Tmax_C,
  Tmin_C,
  windSp_ms,
  Relhum_frac,
  swrad_wm2,
  lwrad_wm2,
  lat_deg,
  slope = 0,
  aspect = 0,
  tempHt = 1,
  windHt = 2,
  groundAlbedo = 0.25,
  SurfEmissiv = 0.95,
  forest = 0,
  startingSnowDepth_m = 0,
  startingSnowDensity_kg_m3 = 550,
  Train = 10,
  Tsnow = 0,
  Max_SnowDensity_kg_m3 = 550
)
```

Arguments

Date	Date(YYYY-MM-DD)
precip_mm	Precipitation in mm/day
Tmax_C	Maximum temperature in degree Celsius
Tmin_C	Minimum temperature in degree Celsius
windSp_ms	Wind speed in m/sec
Relhum_frac	Relative humidity in fraction
swrad_wm2	Incoming shortwave (solar) radiation in W/m2
lwrad_wm2	Incoming longwave radiation in W/m2
lat_deg	Latitude in degree
slope	Slope factor
aspect	Ground aspect [rad from north]
tempHt	Height of temperature station

windHt	Height of wind station
groundAlbedo	Ground albedo (default=0.25)
SurfEmissiv	Surface Emissivity
forest	Forest ratio
startingSnowDepth_m	Initial value of snow depth in meter
startingSnowDensity_kg_m3	Initial value of snow density in kg/m3
Train	Upper temperature of S curve: precipitation is all rain if temperature >= Train
Tsnow	Lower temperature of S curve: precipitation is all snow if temperature <= Tsnow
Max_SnowDensity_kg_m3	Maximum snow density in kg/m3, default=550 from Voordendag et al.(2021)

Details

A modified version of PBSM-S-Curve which simulates snow processes with seven hydroclimate variables. All of hydroclimate variables are from new version of the hybrid climate dataset.

Value

Time series of snow simulation

SnowMelt_PBSM_org	<i>PBSM snow</i>
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Description

Original PBSM snow model that requires only precipitation and temperature

Usage

```
SnowMelt_PBSM_org(
  Date,
  precip_mm,
  Tmax_C,
  Tmin_C,
  lat_deg,
  slope = 0,
  aspect = 0,
  tempHt = 1,
  windHt = 2,
  groundAlbedo = 0.25,
  SurfEmissiv = 0.95,
  windSp = 2,
  forest = 0,
  startingSnowDepth_m = 0,
  startingSnowDensity_kg_m3 = 450
)
```

Arguments

Date	Date (YYYY-MM-DD)
precip_mm	Precipitation in mm/day
Tmax_C	Maximum temperature in degree Celsius
Tmin_C	Minimum temperature in degree Celsius
lat_deg	Latitude in degree
slope	Slope factor
aspect	Ground aspect [rad from north]
tempHt	Height of temperature station
windHt	Height of wind station
groundAlbedo	Ground albedo (default=0.25)
SurfEmissiv	Surface Emissivity
windSp	Wind speed in m/sec (default=2.0)
forest	Forest ratio
startingSnowDepth_m	Initial value of snow depth in meter
startingSnowDensity_kg_m3	Initial value of snow density in kg/m3

Value

Time series of snow simulation

References

- Todd Walter, M., Brooks, E.S., McCool, D.K., King, L.G., Molnau, M., Boll, J., 2005. Process-based snowmelt modeling: does it require more input data than temperature-index modeling? *Journal of Hydrology* 300, 65–75. <https://doi.org/10.1016/j.jhydrol.2004.05.002>
- Fuka, D.R., Easton, Z.M., Brooks, E.S., Boll, J., Steenhuis, T.S., Walter, M.T., 2012. A Simple Process-Based Snowmelt Routine to Model Spatially Distributed Snow Depth and Snowmelt in the SWAT Model, *JAWRA Journal of the American Water Resources Association* 48, 1151–1161. <https://doi.org/10.1111/j.1752-1688.2012.00680.x>

SnowMelt_S_Curve

PBSM with S-curve

Description

Original PBSM has been Modified by incorporating a S-shape curve method that separates rain and snow by a S-curve Added ice rain in snowpack if snowpack temperature is below 0 and rain is on the snowpack.

Usage

```

SnowMelt_S_Curve(
  Date,
  precip_mm,
  Tmax_C,
  Tmin_C,
  lat_deg,
  slope = 0,
  aspect = 0,
  tempHt = 1,
  windHt = 2,
  groundAlbedo = 0.25,
  SurfEmissiv = 0.95,
  windSp = 2,
  forest = 0,
  startingSnowDepth_m = 0,
  startingSnowDensity_kg_m3 = 550,
  Train = 10,
  Tsnow = 0,
  Max_SnowDensity_kg_m3 = 550
)

```

Arguments

Date	Date (YYYY-MM-DD)
precip_mm	Precipitation in mm/day
Tmax_C	Maximum temperature in degree Celsius
Tmin_C	Minimum temperature in degree Celsius
lat_deg	Latitude in degree
slope	Slope factor
aspect	Ground aspect [rad from north]
tempHt	Height of temperature station
windHt	Height of wind station
groundAlbedo	Ground albedo (default=0.25)
SurfEmissiv	Surface Emissivity
windSp	Wind speed in m/sec (default=2.0)
forest	Forest ratio
startingSnowDepth_m	Initial value of snow depth in meter
startingSnowDensity_kg_m3	Initial value of snow density in kg/m3
Train	Upper temperature of S curve: precipitation is all rain if temperature >= Train
Tsnow	Lower temperature of S curve: precipitation is all snow if temperature <= Tsnow
Max_SnowDensity_kg_m3	Maximum snow density in kg/m3, default=550 from Voordendag et al.(2021)

Details

Added two arguments (Train and Tsnow) that are minimum temperature (Train) for all precipitation and maximum temperature (Tsnow) for all snow. Author: Hyung Eum, AEP Date: March 2021

Value

Time series of snow simulation

References

Todd Walter, M., Brooks, E.S., McCool, D.K., King, L.G., Molnau, M., Boll, J., 2005. Process-based snowmelt modeling: does it require more input data than temperature-index modeling? Journal of Hydrology 300, 65–75. <https://doi.org/10.1016/j.jhydrol.2004.05.002>

Fuka, D.R., Easton, Z.M., Brooks, E.S., Boll, J., Steenhuis, T.S., Walter, M.T., 2012. A Simple Process-Based Snowmelt Routine to Model Spatially Distributed Snow Depth and Snowmelt in the SWAT Model. JAWRA Journal of the American Water Resources Association 48, 1151–1161. <https://doi.org/10.1111/j.1752-1688.2012.00680.x>

Stefan W. Kienzle (2008). A new temperature based method to separate rain and snow. Hydrological processes, 22, 5067-5085.

SnowMelt_S_Curve_Sublim

PBSM with S-curve and Sublimation

Description

Original PBSM has been Modified by incorporating a S-shape curve method that separates rain and snow by a S-curve Added ice rain in snowpack if snowpack temperature is below 0 and rain is on the snowpack. Added Sublimation physics (Mass transfer based on Latent Heat Flux).

Usage

```
SnowMelt_S_Curve_Sublim(
  Date,
  precip_mm,
  Tmax_C,
  Tmin_C,
  lat_deg,
  slope = 0,
  aspect = 0,
  tempHt = 1,
  windHt = 2,
  groundAlbedo = 0.25,
  SurfEmissiv = 0.95,
  windSp = 2,
  forest = 0,
  startingSnowDepth_m = 0,
  startingSnowDensity_kg_m3 = 550,
  Train = 10,
  Tsnow = 0,
  Max_SnowDensity_kg_m3 = 550
)
```

Arguments

Date	Date (YYYY-MM-DD)
precip_mm	Precipitation in mm/day
Tmax_C	Maximum temperature in degree Celsius
Tmin_C	Minimum temperature in degree Celsius
lat_deg	Latitude in degree
slope	Slope factor
aspect	Ground aspect [rad from north]
tempHt	Height of temperature station
windHt	Height of wind station
groundAlbedo	Ground albedo (default=0.25)
SurfEmissiv	Surface Emissivity
windSp	Wind speed in m/sec (default=2.0)
forest	Forest ratio
startingSnowDepth_m	Initial value of snow depth in meter
startingSnowDensity_kg_m3	Initial value of snow density in kg/m3
Train	Upper temperature of S curve: precipitation is all rain if temperature $\geq$ Train
Tsnow	Lower temperature of S curve: precipitation is all snow if temperature $\leq$ Tsnow
Max_SnowDensity_kg_m3	Maximum snow density in kg/m3, default=550 from Voordendag et al.(2021)

Details

Added two arguments (Train and Tsnow) that are minimum temperature (Train) for all precipitation and maximum temperature (Tsnow) for all snow. Author: Hyung Eum, AEP Date: March 2021 (Modified for Sublimation Feb 2026)

Value

Time series of snow simulation including sublimation

Solar

Solar radiation

Description

Estimate solar radiation

Usage

```
Solar(
  lat,
  Jday,
  Tx,
  Tn,
  albedo = 0.2,
  forest = 0,
  slope = 0,
  aspect = 0,
  units = "kJm2d",
  latUnits = "unknown",
  printWarn = TRUE
)
```

Arguments

lat	Latitude: degree or rad
Jday	Julian date or day of the year [day]
Tx	Maximum temperature in degree Celcius
Tn	Minimum temperature in degree Celcius
albedo	Surface albedo (default=0.2)
forest	Forest ratio
slope	Slope
aspect	Ground aspect [rad from north]
units	Radiation unit: Wm2 or kJm2d (default=kJm2d)
latUnits	Latitude unit (default=unknown)
printWarn	option for printing warning (default=TRUE)

Value

Estimated solar radiation

solarangle	<i>Solar Angle</i>
------------	--------------------

Description

Angle of solar inclination from horizontal at solar noon [rad]

Usage

```
solarangle(lat, Jday)
```

Arguments

lat	latitdue [rad]
Jday	Julian date or day of the year [day]

Value

Solar angle in radian

transmissivity	<i>Transmissivity</i>
----------------	-----------------------

Description

Fraction of direct solar radiation passing through the atmosphere based on the Bristow-Campbell eqn

Usage

transmissivity(Tx, Tn, A = 0.75, C = 2.4, opt = "1day", JD = NULL)

Arguments

- | | |
|-----|---|
| Tx | maximum daily temperature [C] |
| Tn | minimum daily temperature [C] |
| A | Empirical coefficient (default=0.75) |
| C | Empirical coefficient (default=2.4) |
| opt | average diurnal temperature methods: 1day (default), 2day and missingdays |
| JD | Julian Day (default=NULL) |

Value

Atmospheric transmissivity

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